

- 2 (a) (i) When the coil rotates, there is a rate of change magnetic flux linking to the circuit. According to Faraday's law of electromagnetic induction, an emf is induced in a circuit when there is a change in its magnetic flux linkage. The magnitude of the induced emf is directly proportional to the rate of change of magnetic flux linkage.

Note: It was expected that candidates would indicate that the sides of the rotating coil would cut the magnetic flux of the magnet.

(ii) 1. The magnetic flux density within which the coil is located.

2. The frequency of the rotation of the coil.

(iii) As the coil rotates at a constant angular frequency ω , the angle $\theta = \omega t$ changes at constant rate. Hence the magnetic flux density perpendicular to the coil varies sinusoidally, and the rate at which it is changing (the emf) is also sinusoidal.

Note: It was expected that that a reference would be made to the dependence of the emf on the angle between the coil and the field and that this angle varies sinusoidally with time.

(b) (i) For sinusoidally varying waveforms, the peak emf $E_0 = \sqrt{2}E_{rms} = \sqrt{2}(72) = 102 \text{ V}$

$$\begin{aligned} \text{(ii)} \quad \frac{V_s}{V_p} &= \frac{N_s}{N_p} \\ \frac{V_s}{72} &= \frac{600}{30} \\ V_s &= 1440 \text{ V} \end{aligned}$$

$$\text{(iii)} \quad I_{rms} = \frac{V_{rms}}{R} = \frac{1440}{160} = 9.0 \text{ A}$$

$$\text{(iv)} \quad \langle P \rangle = I_{rms} V_{rms} = (9.0)(1440) = 1.3 \times 10^4 \text{ W}$$

(v) For ideal transformer, input power = output power

$$\text{At the input, } I_{rms} = \frac{\langle P_{mean} \rangle}{V_{rms}} = \frac{1.3 \times 10^4}{72} = 180$$

3 (a) A stationary wave is a wave with a waveform that does not advance and there is no